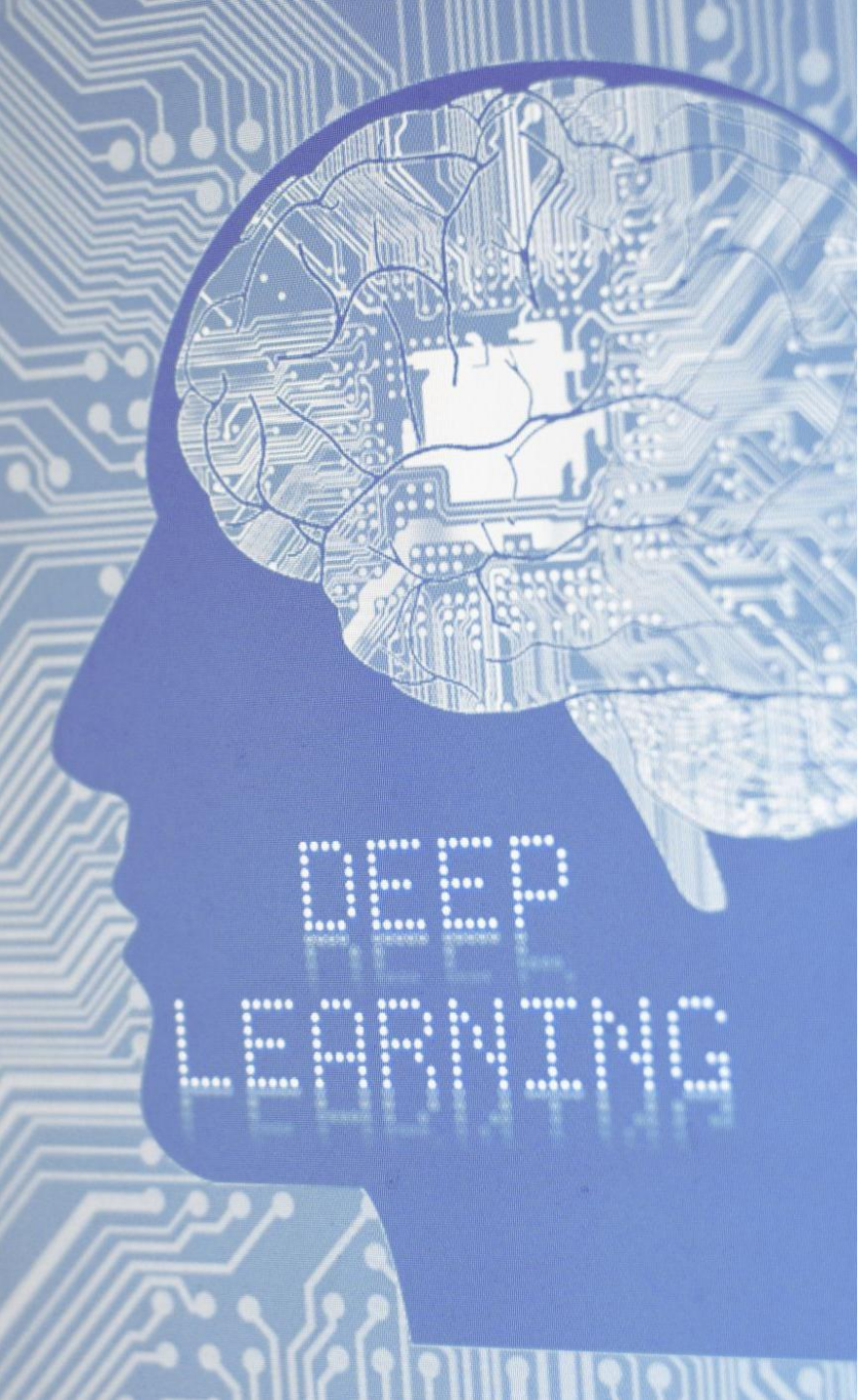
Spring of Code

Artificial Intelligence

(Machine Learning & Deep Learning)

Instructor
Asad Ullah
- *Research Assistant* (DIP Lab)

Duration
03 Months
(Feb – April)



Day 01 – Deep Learning
(Artificial Neural Networks)

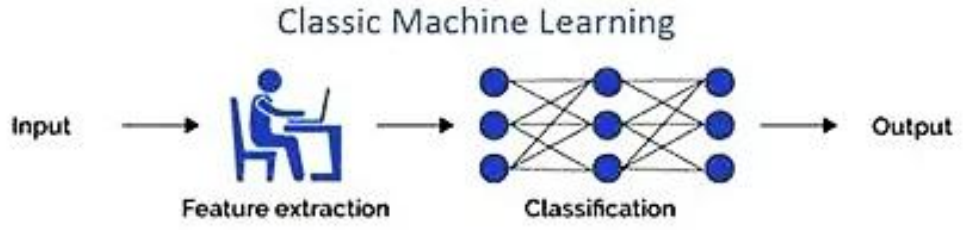
Objectives:

- Introduction to Deep Learning
- Artificial Neural Networks
- Activation Functions

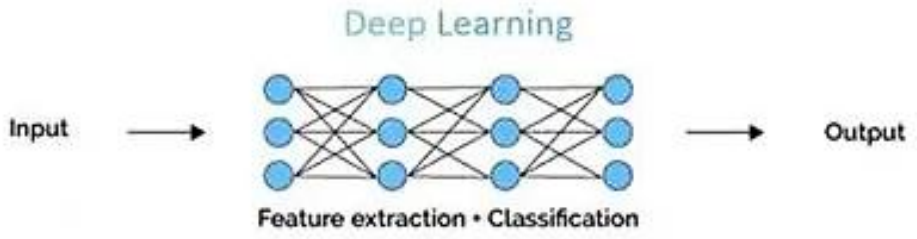
Deep Learning

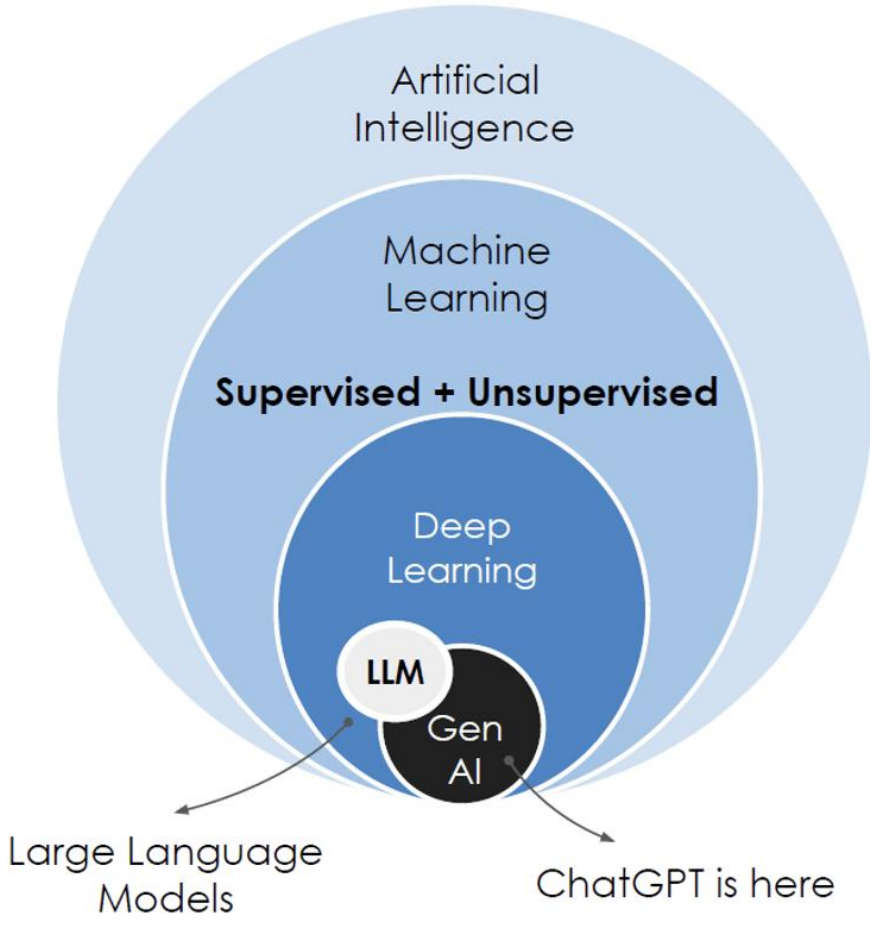
- Deep Learning is a subset of machine learning that uses neural networks to learn representations of data.
- The fundamental unit of such networks is a small bundle of computation called an artificial neuron.

Machine Learning:



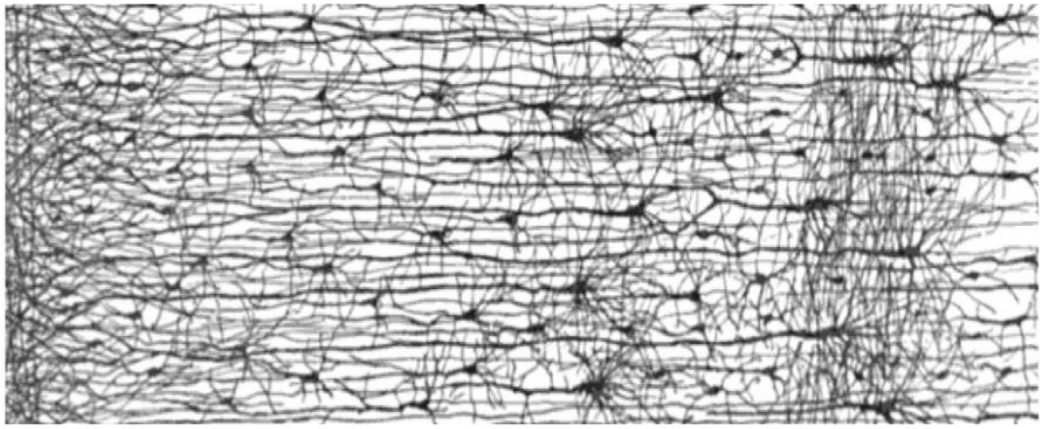
Deep Learning:

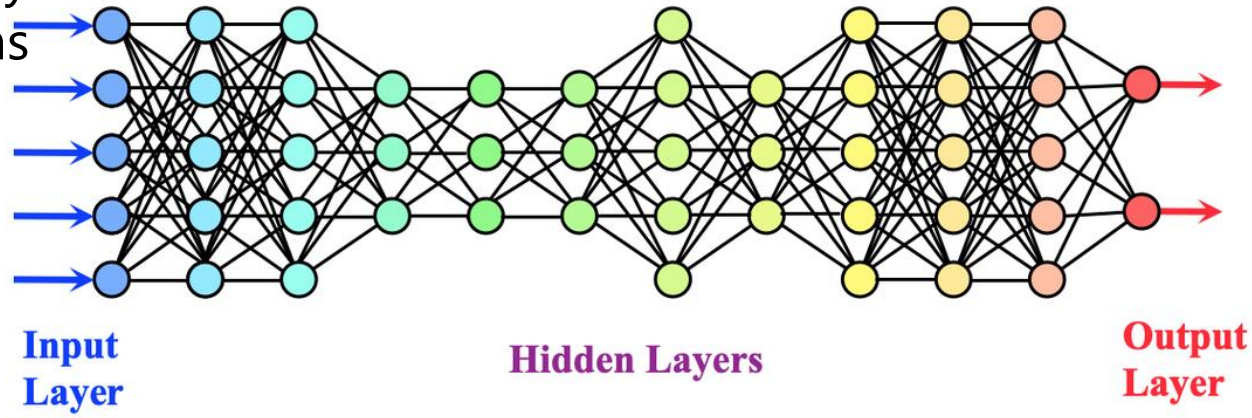


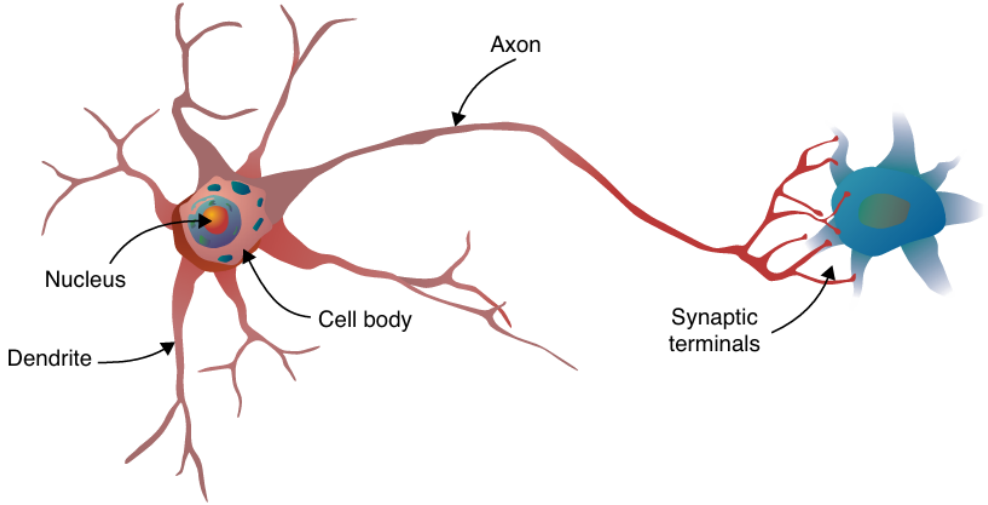


Neural Networks

- A Neural Network is a computational model inspired by the way biological neurons in the human brain process information.
- Composed of interconnected nodes (neurons) organized in layers.
- It learns to map inputs to outputs by adjusting connection strengths (weights)

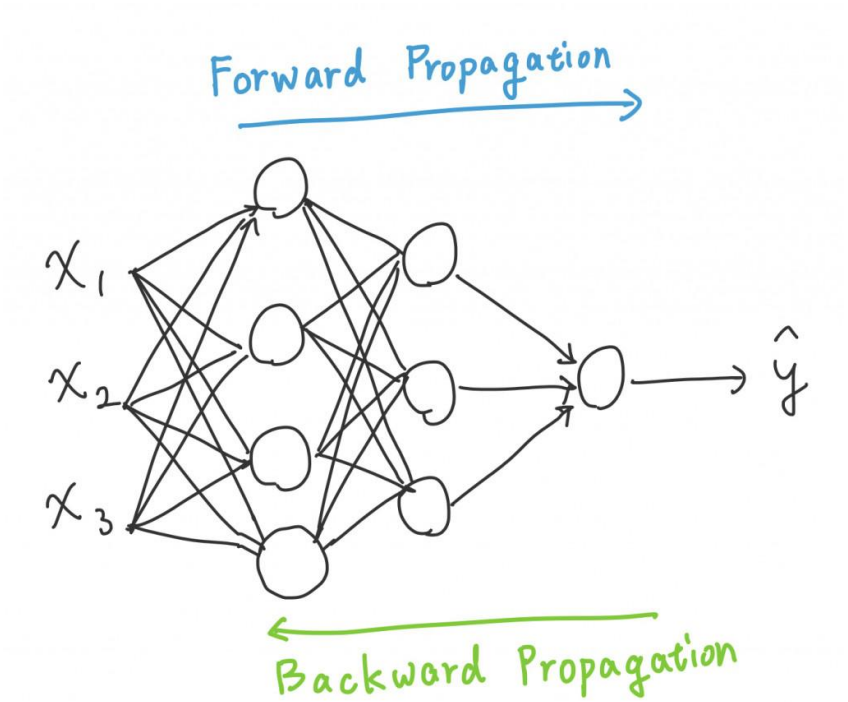


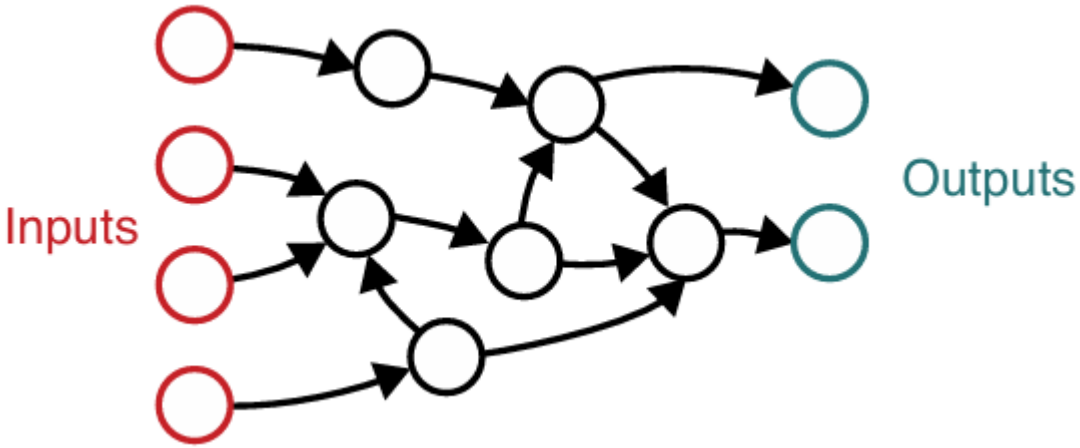




Neural Networks Computation Graphs

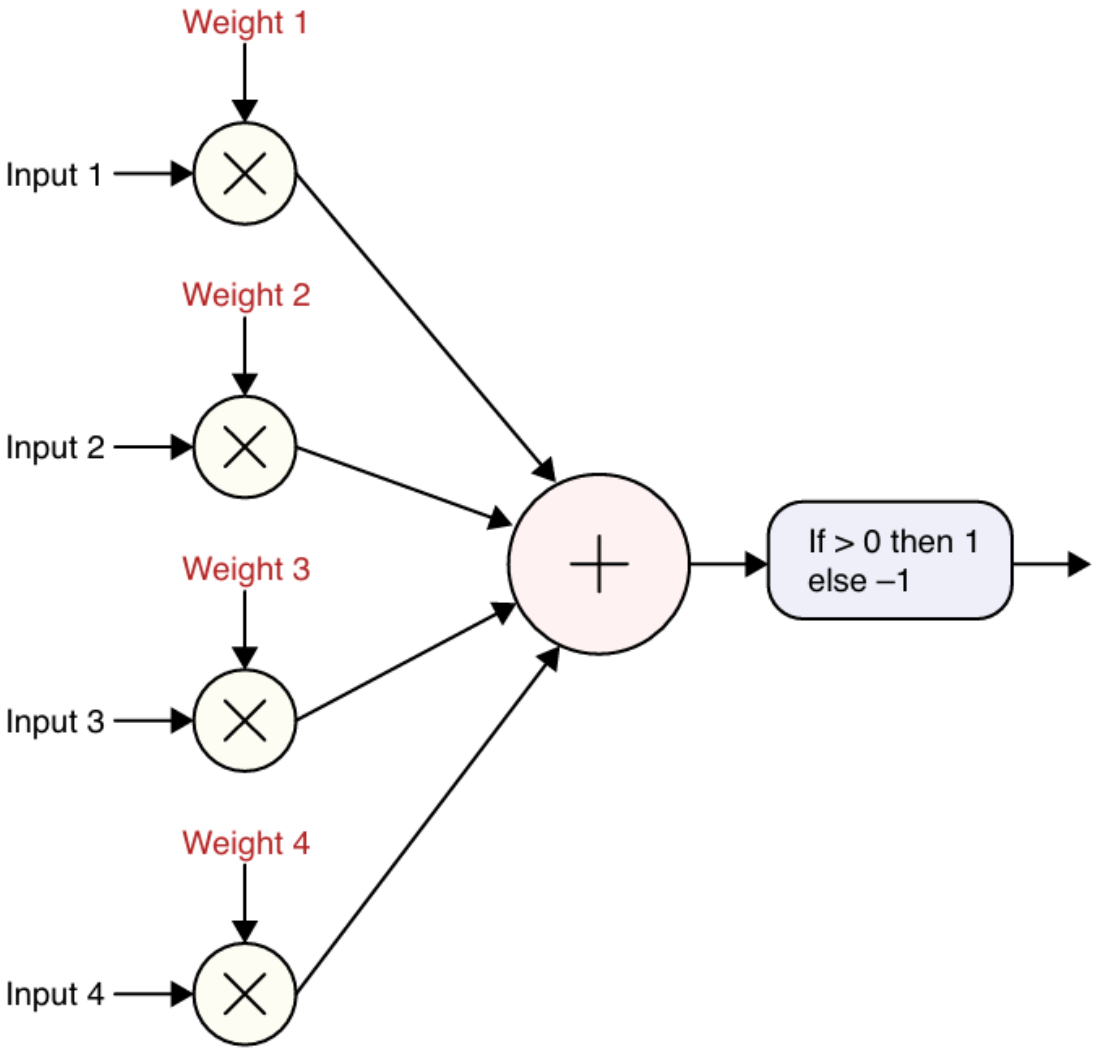
- Neural networks can be represented as directed acyclic graphs (DAGs) where:
  - **Nodes (Vertices)** = Neurons/Units
  - **Edges** = Weighted connections between neurons
  - **Direction** = Flow of information (input to output)
  - **Acyclic** = No loops - information flows forward only



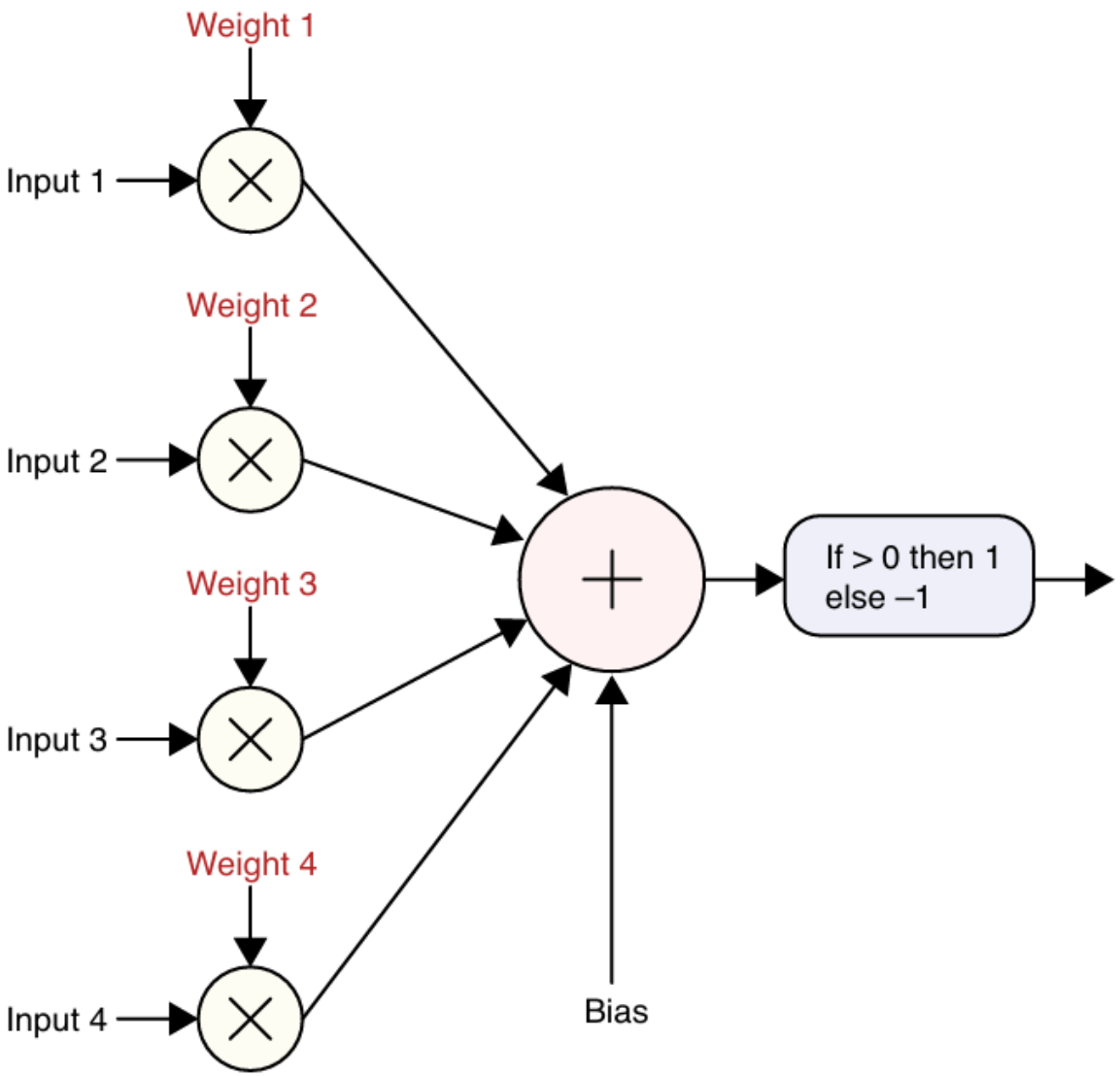


Feed Forward Neural Network

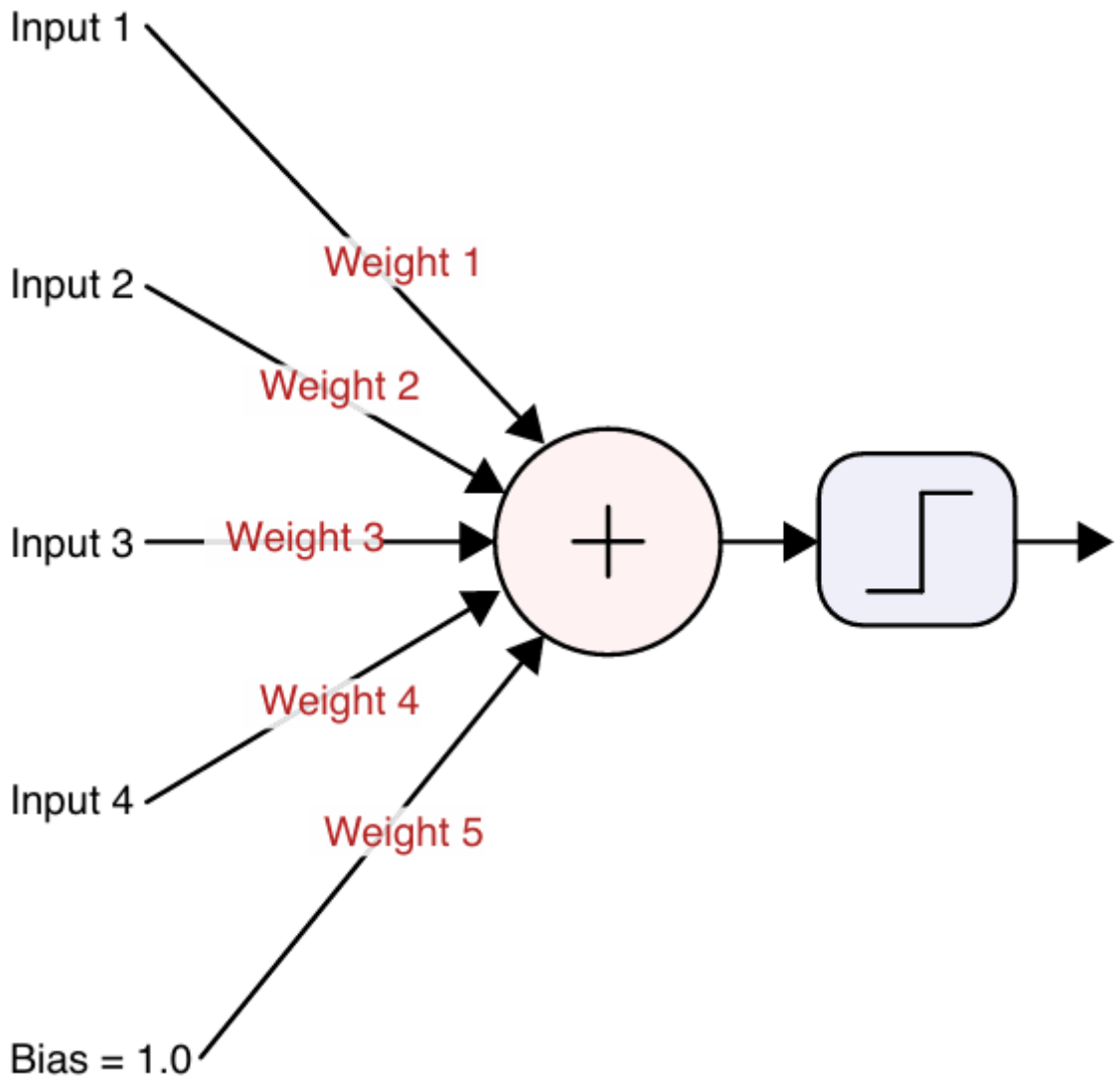
The Perceptron (Original)



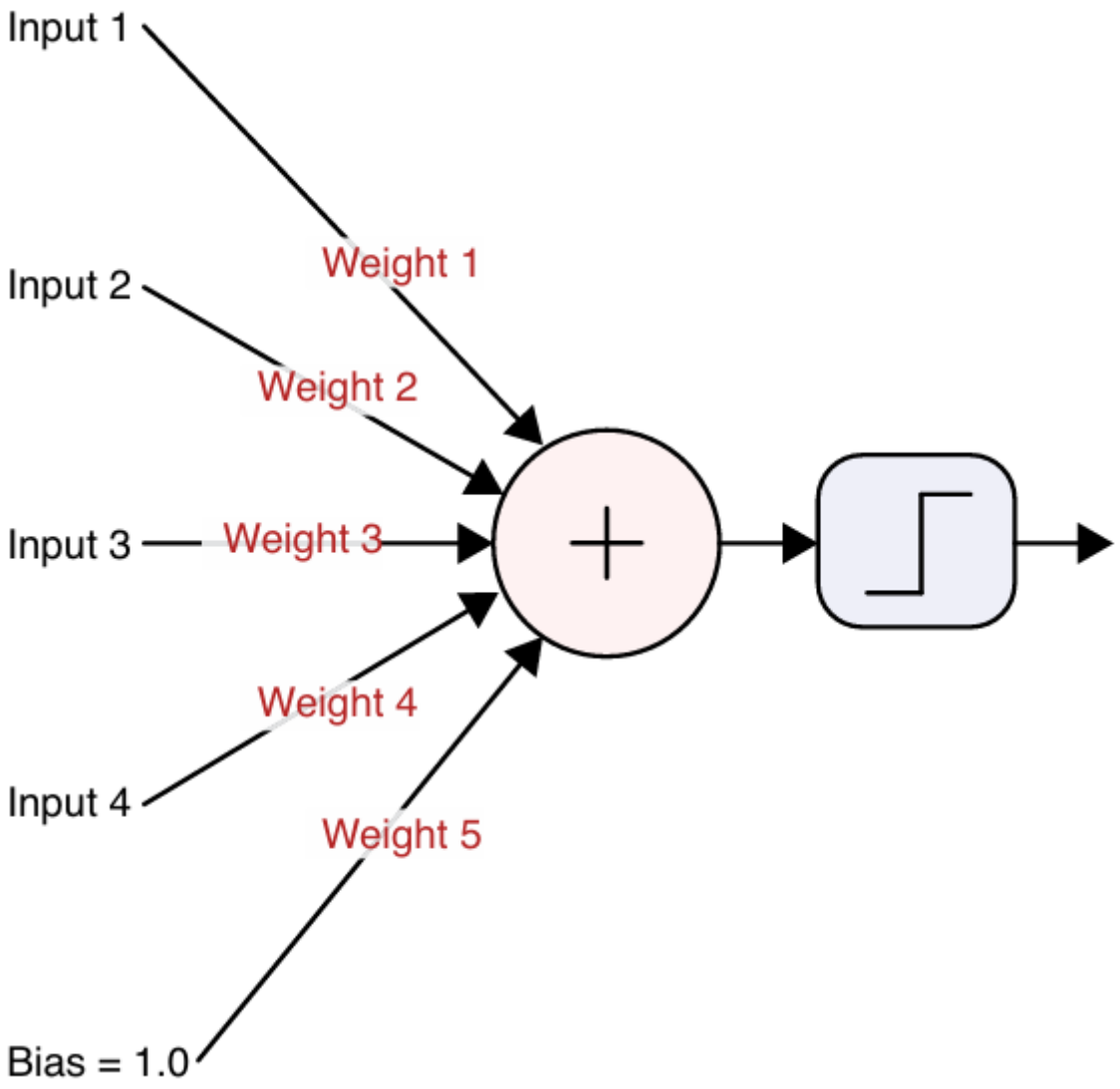
Artificial Neuron (Modern)

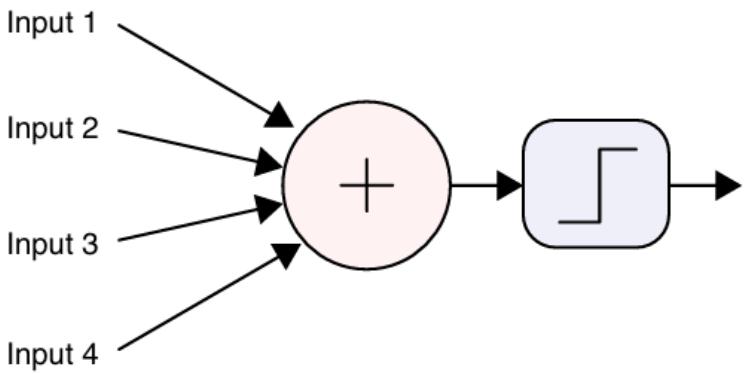


Drawing Artificial Neuron



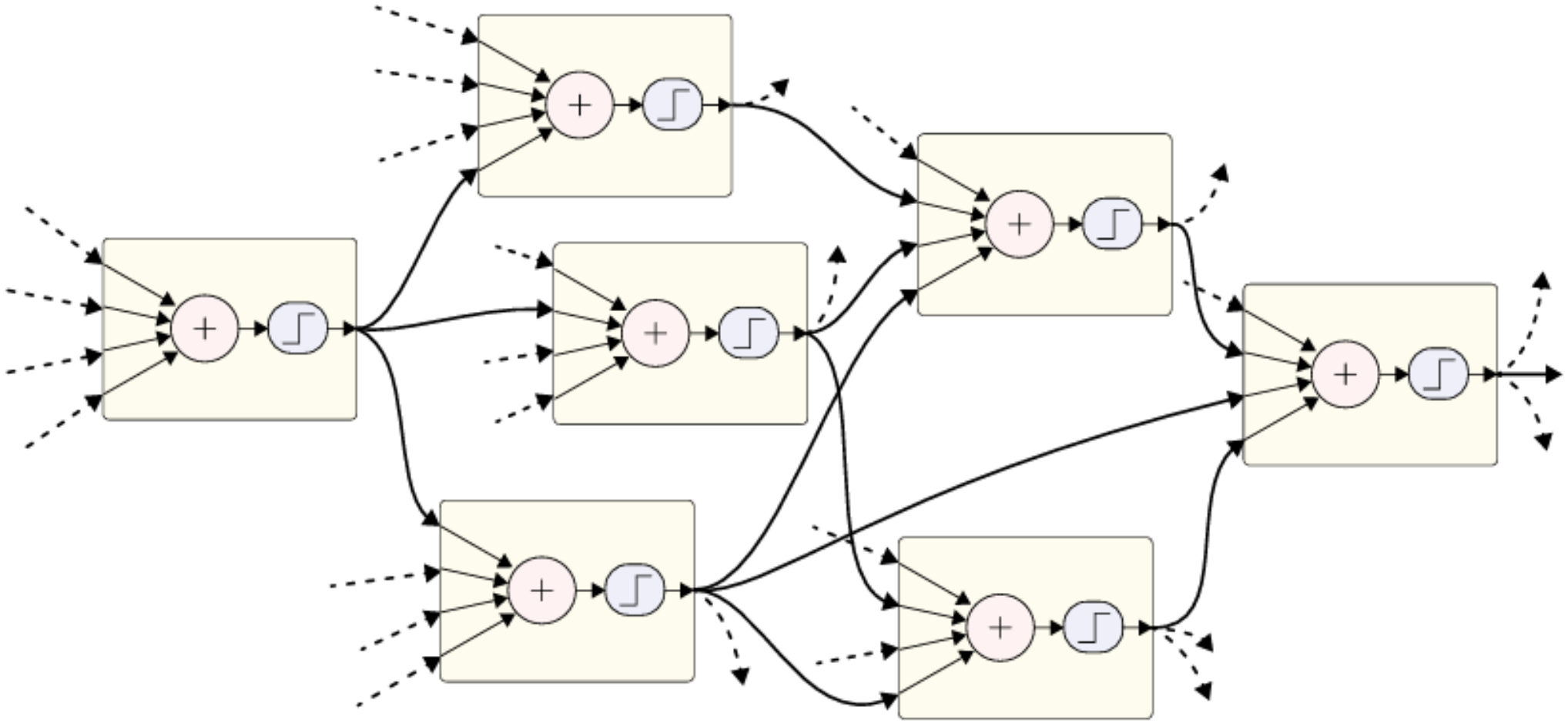
Drawing Artificial Neuron



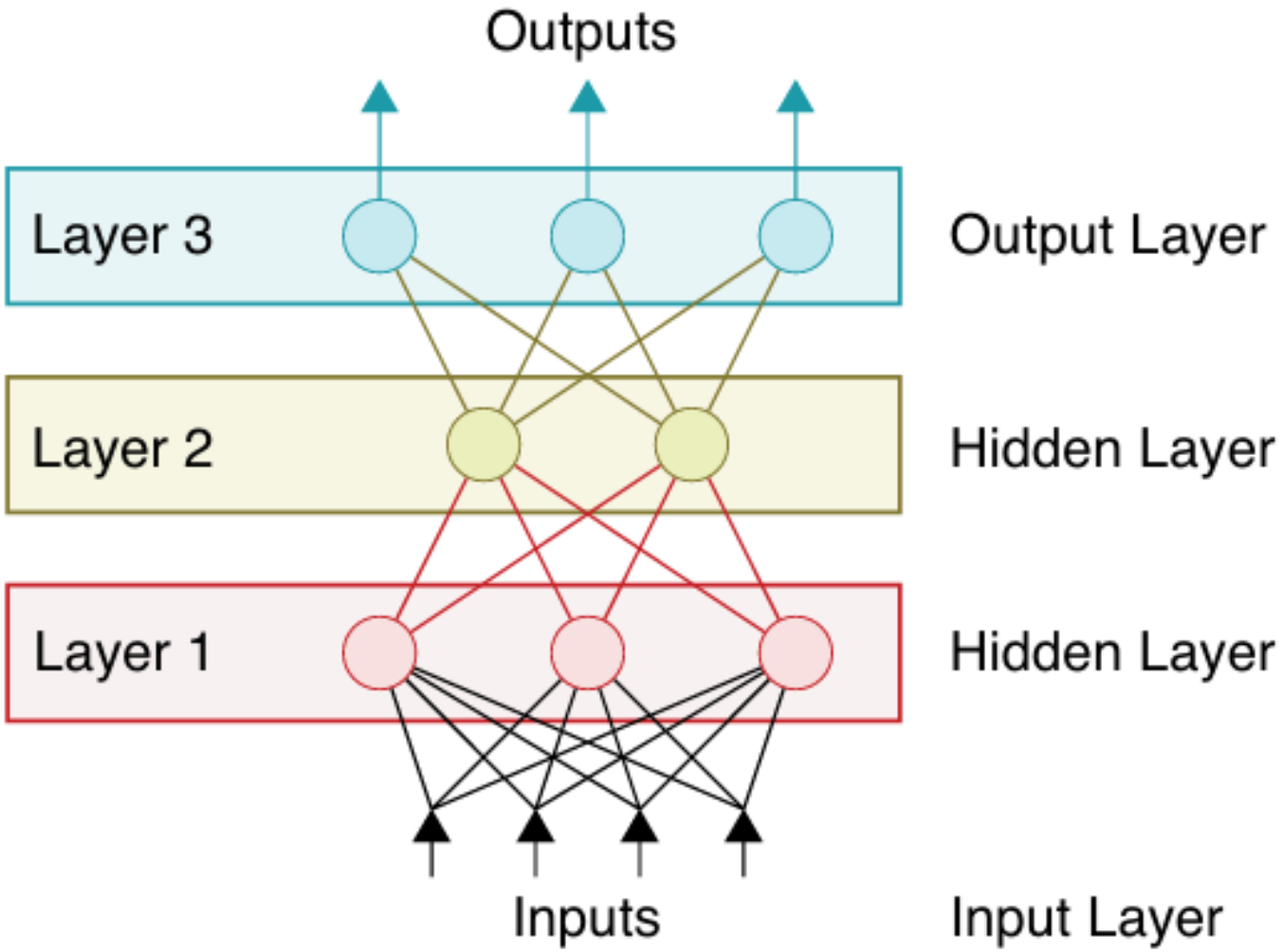


Compact Form

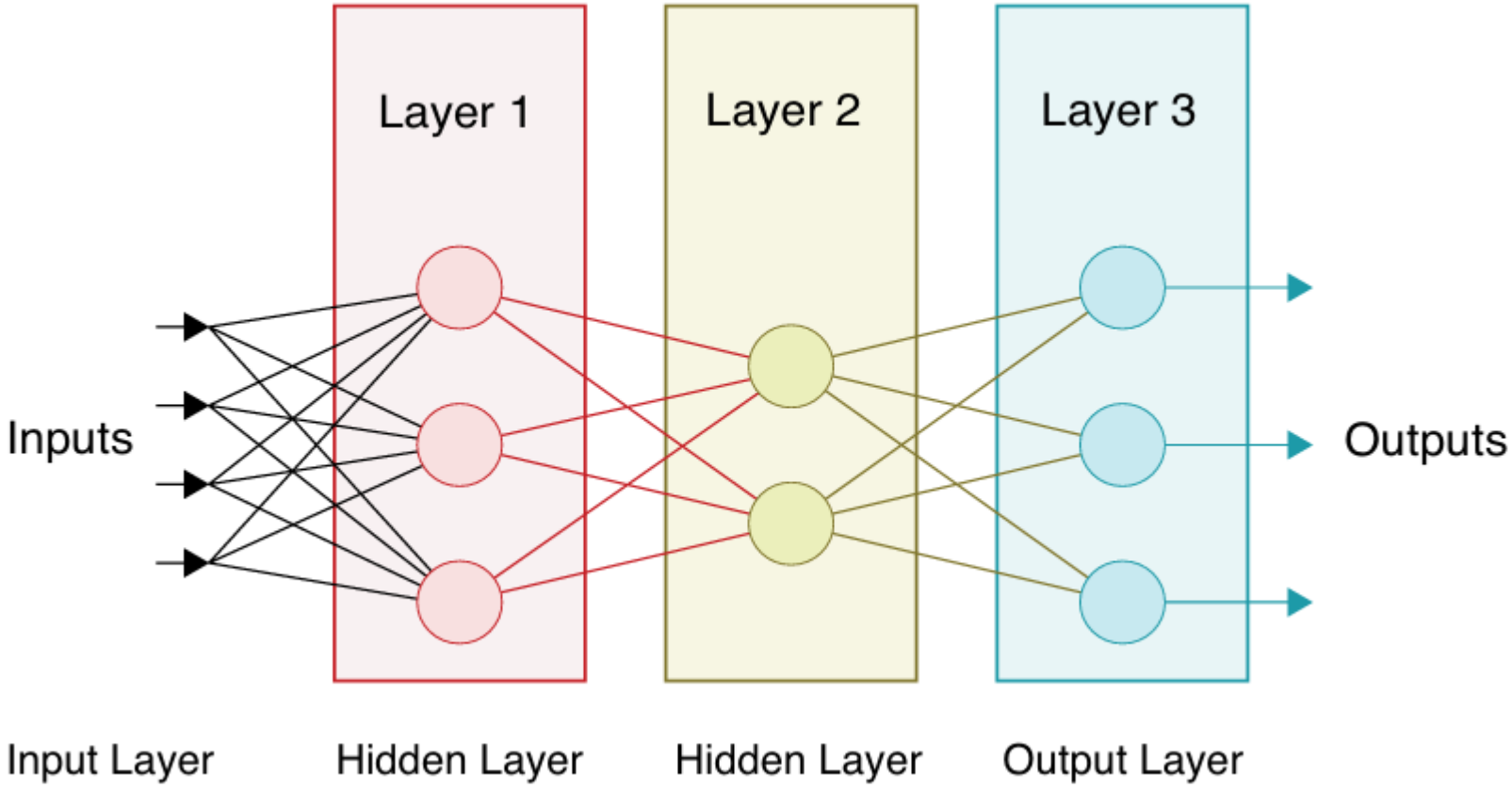
Neural Network



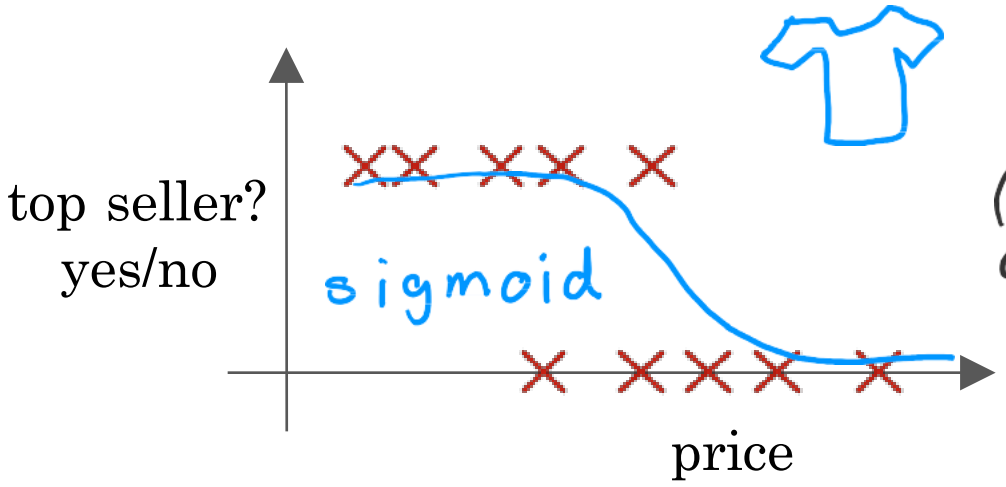
Deep Networks



Deep Networks



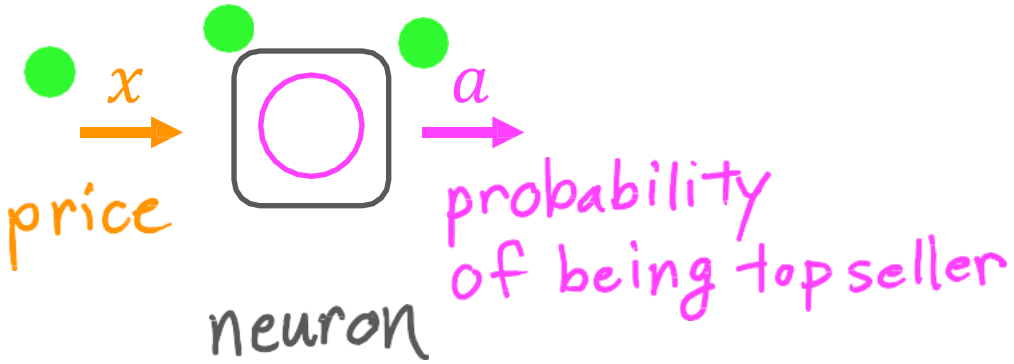
Demand Prediction



$x = price$ input

$$a = f(x) = \frac{1}{1 + e^{-(wx+b)}}$$ output

activation



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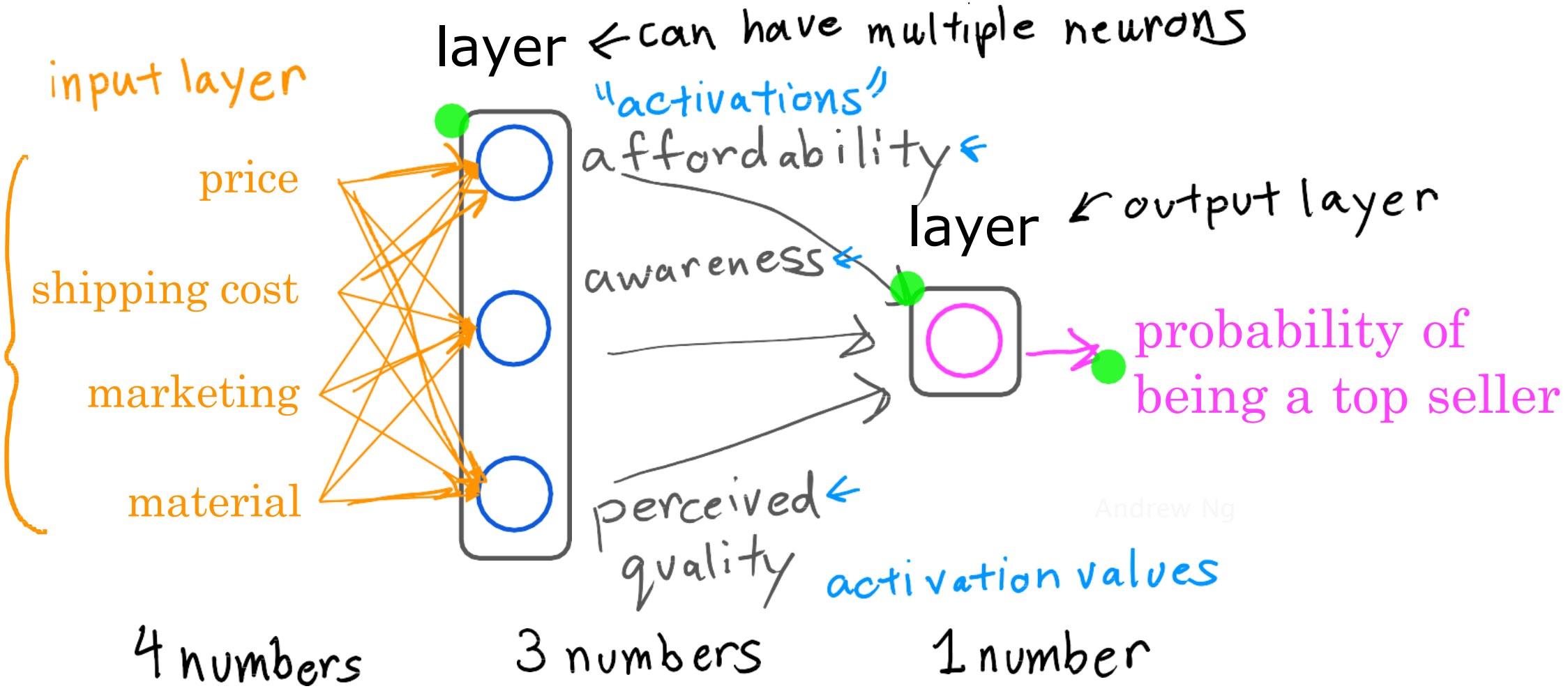
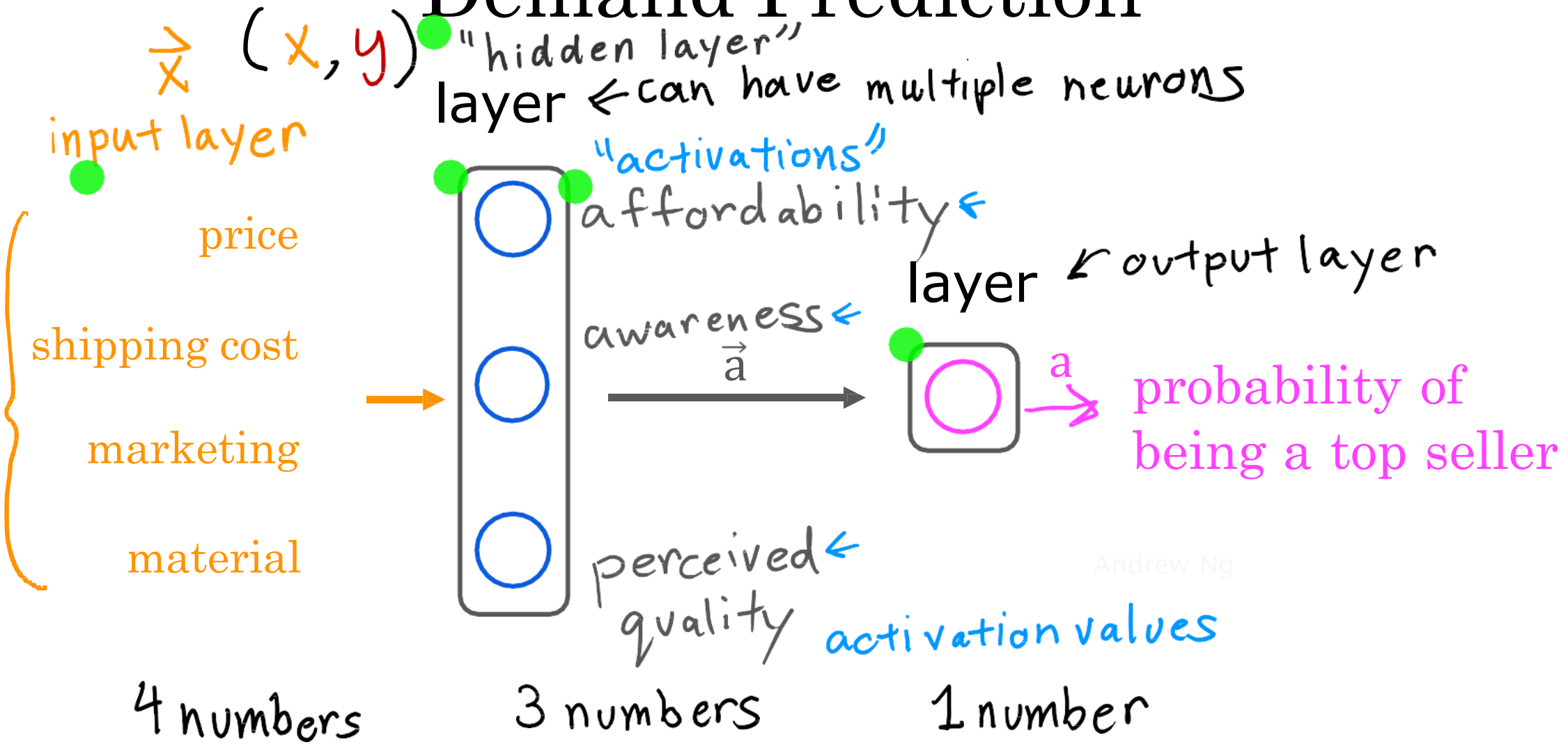
Demand Prediction



Andrew Ng

Demand Prediction



Demand Prediction

feature engineering
x_1, x_2

$\vec{x}$ (x, y)

input layer

price

shipping cost

marketing

material

4 numbers

"hidden layer"
layer ← can have multiple neurons

"activations"

affordability ←

layer ← output layer

awareness ←

$\vec{a}$

perceived quality ←

activation values

3 numbers

1 number

a

probability of being a top seller

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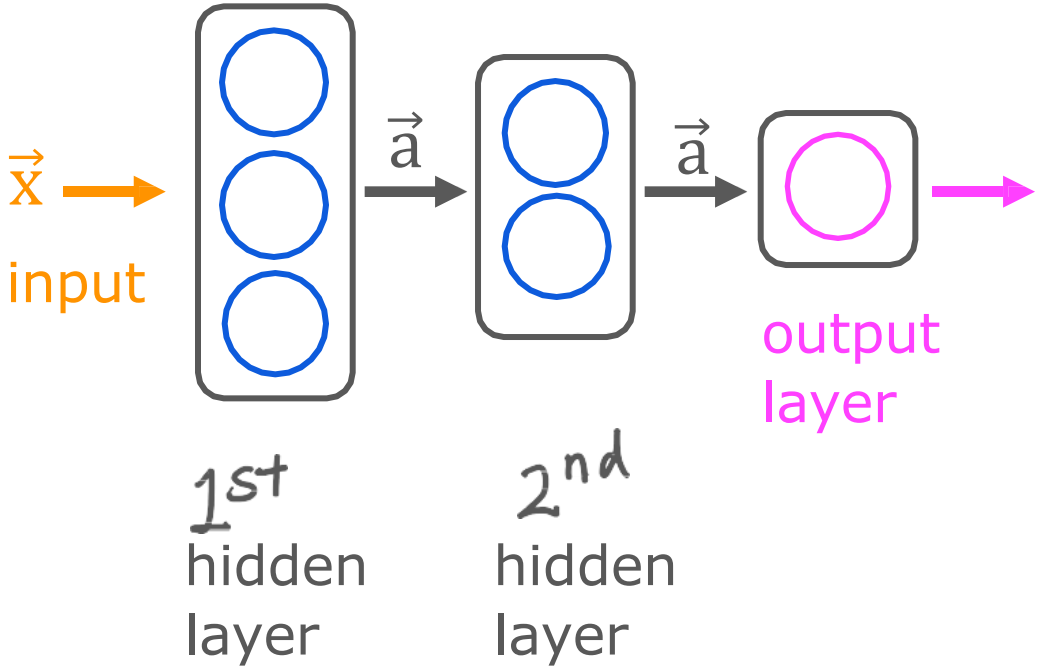
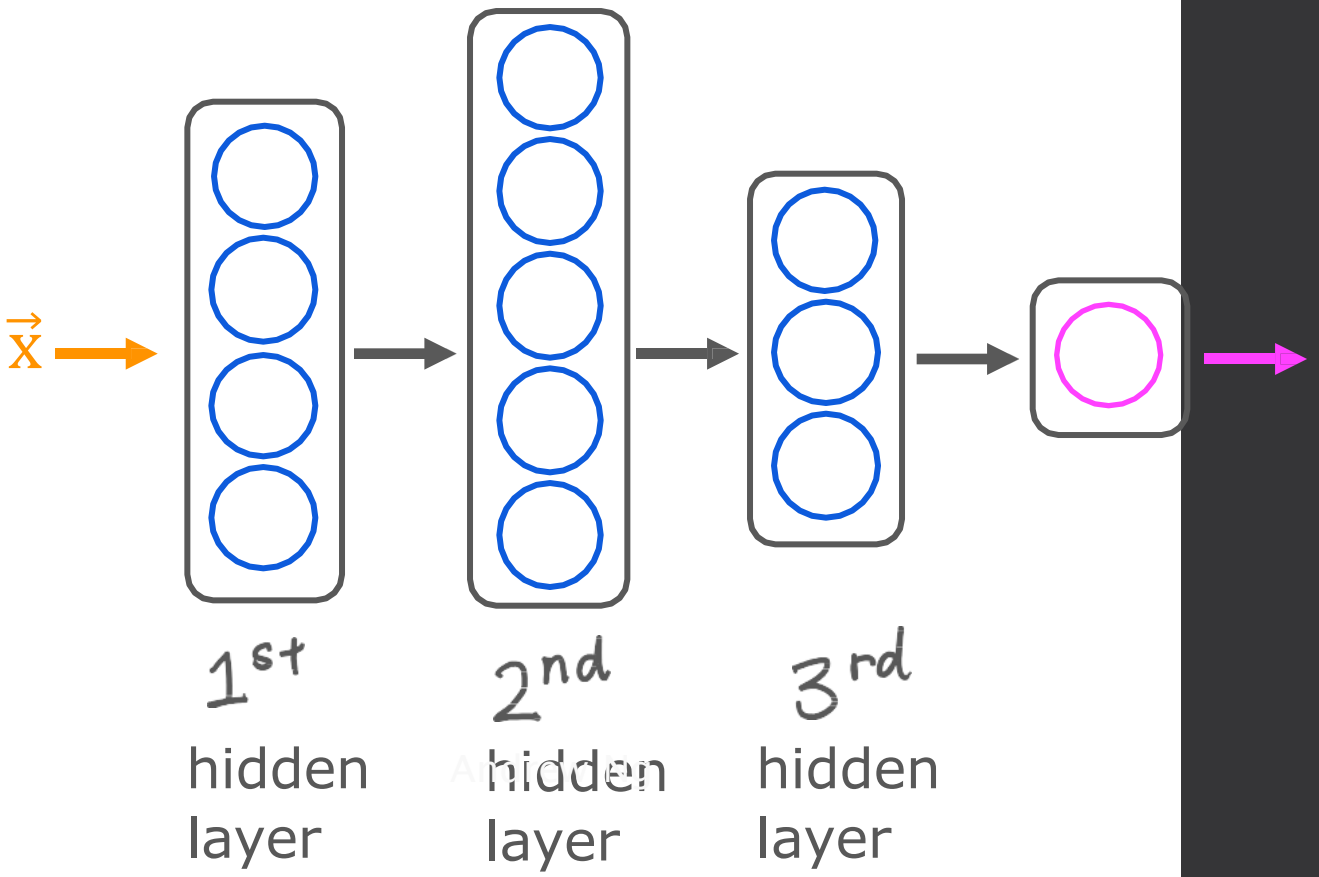
Multiple hidden layers



"multilayer perceptron"



neural network architecture

Face recognition

1000 pixels

1000 pixels

1000 columns

1000 rows

197	185	203	...	...
...	57	64	92	...
⋮				
		...	187	214

$\vec{x} =$

197
185
203
⋮
57
64
92
⋮
187
214

1 million

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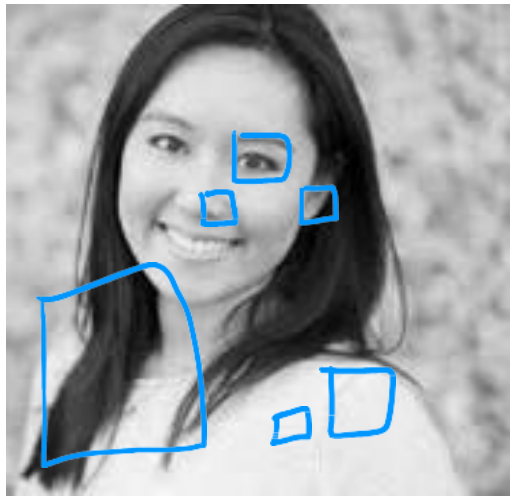
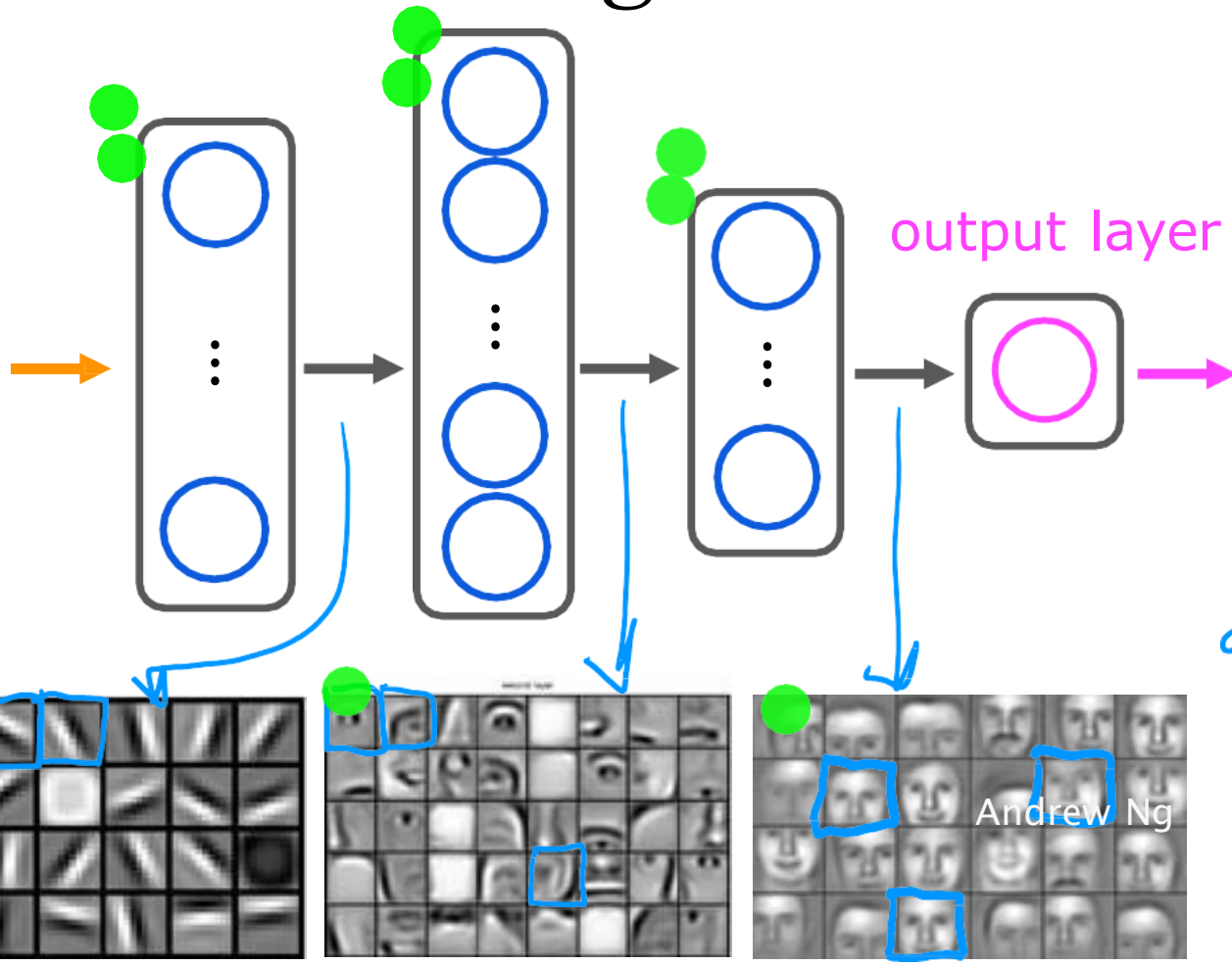
Face recognition



$\vec{x}$

input

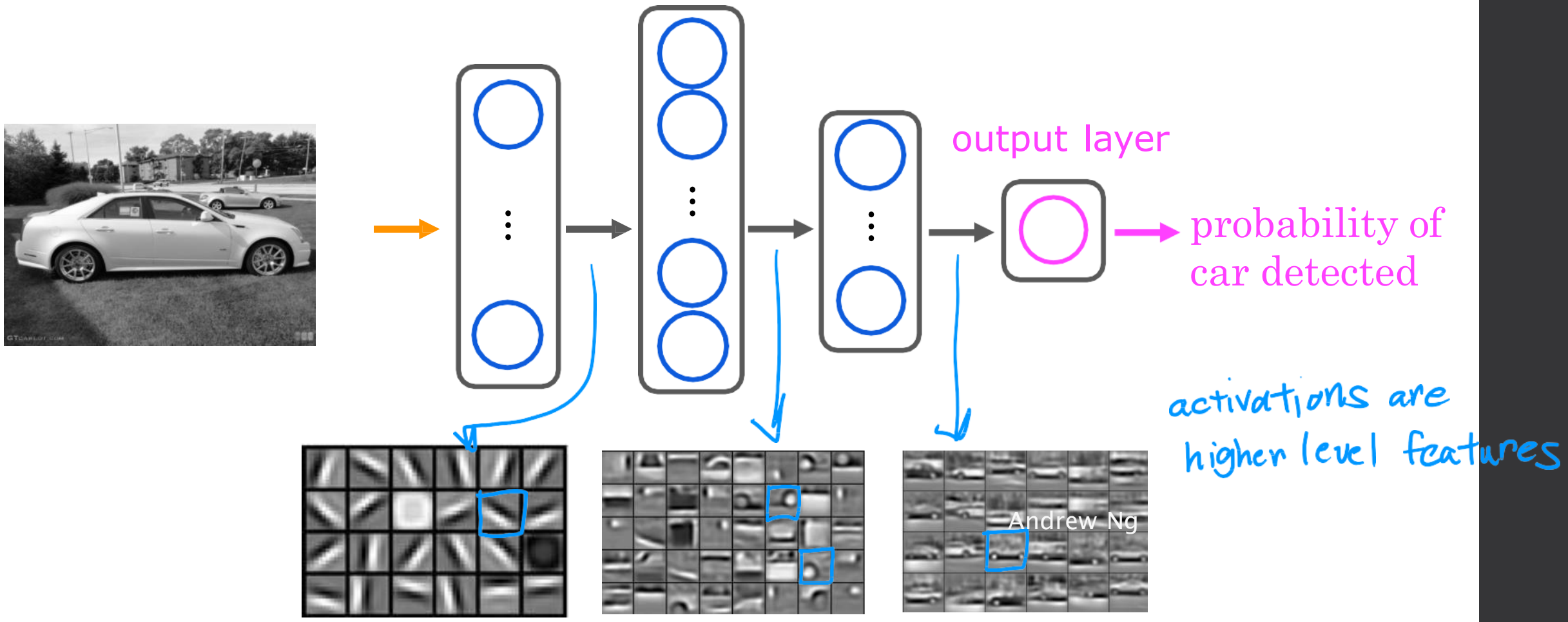


probability of being person 'XYZ'

activations are higher level features

source: Convolutional Deep Belief Networks for Scalable Unsupervised Learning of Hierarchical Representations by Honglak Lee, Roger Grosse, Ranganath Andrew Y. Ng

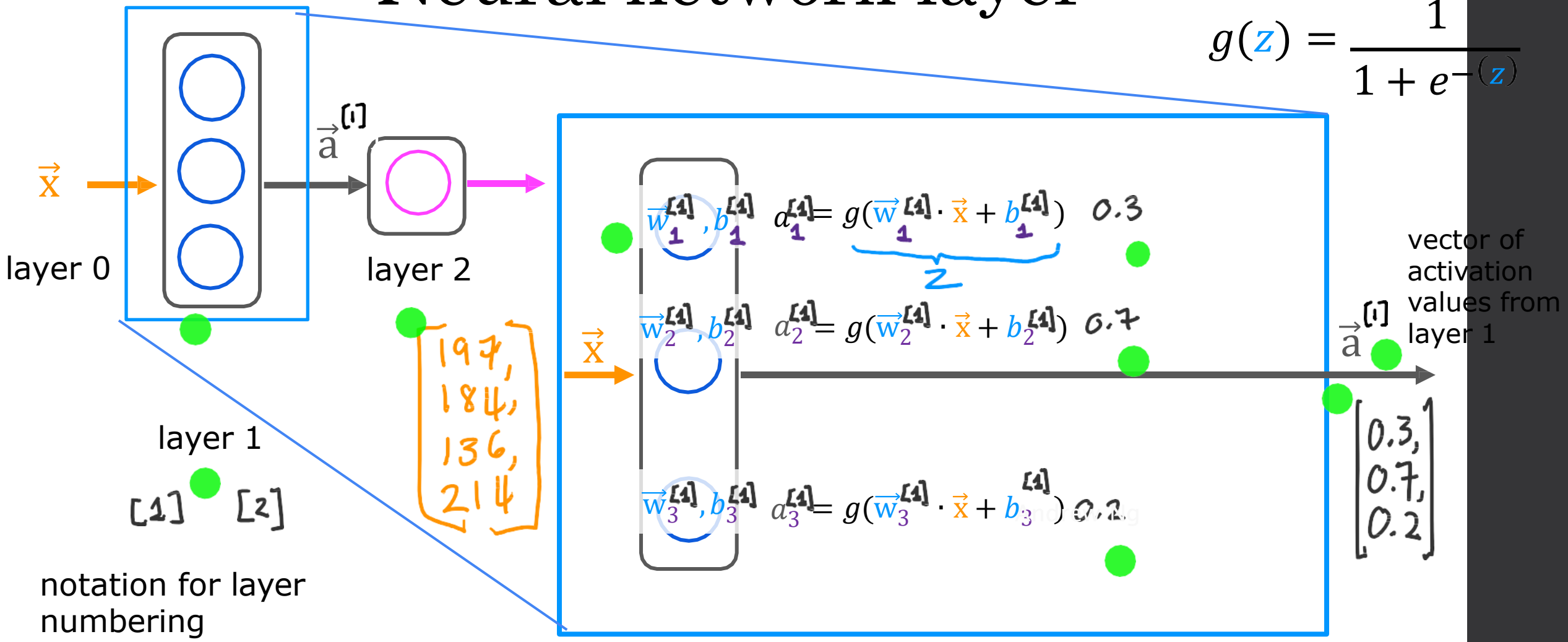
Car classification



source: Convolutional Deep Belief Networks for Scalable Unsupervised Learning of Hierarchical Representations by Honglak Lee, Roger Grosse, Ranganath Andrew Y. Ng

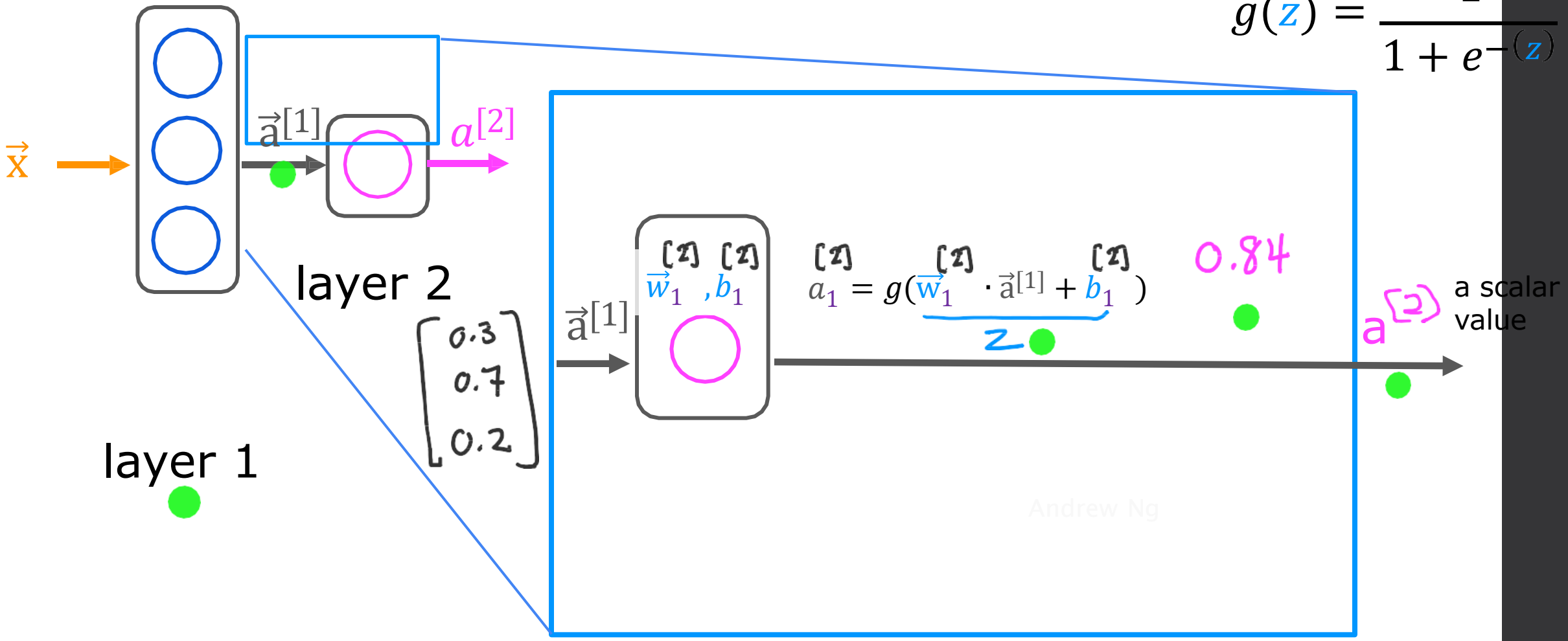
Neural network layer

$$g(z) = \frac{1}{1 + e^{-(z)}}$$

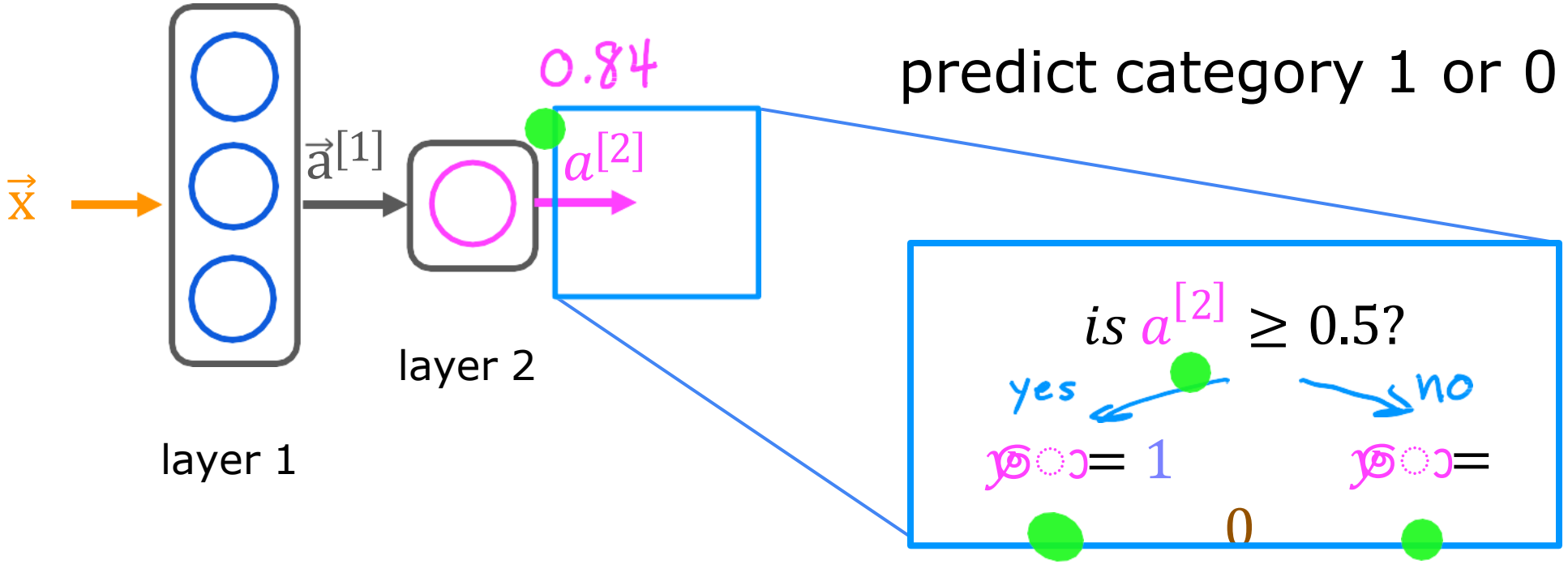


Neural network layer

$$g(z) = \frac{1}{1 + e^{-(z)}}$$



Neural network layer



predict category 1 or 0 (yes/no)

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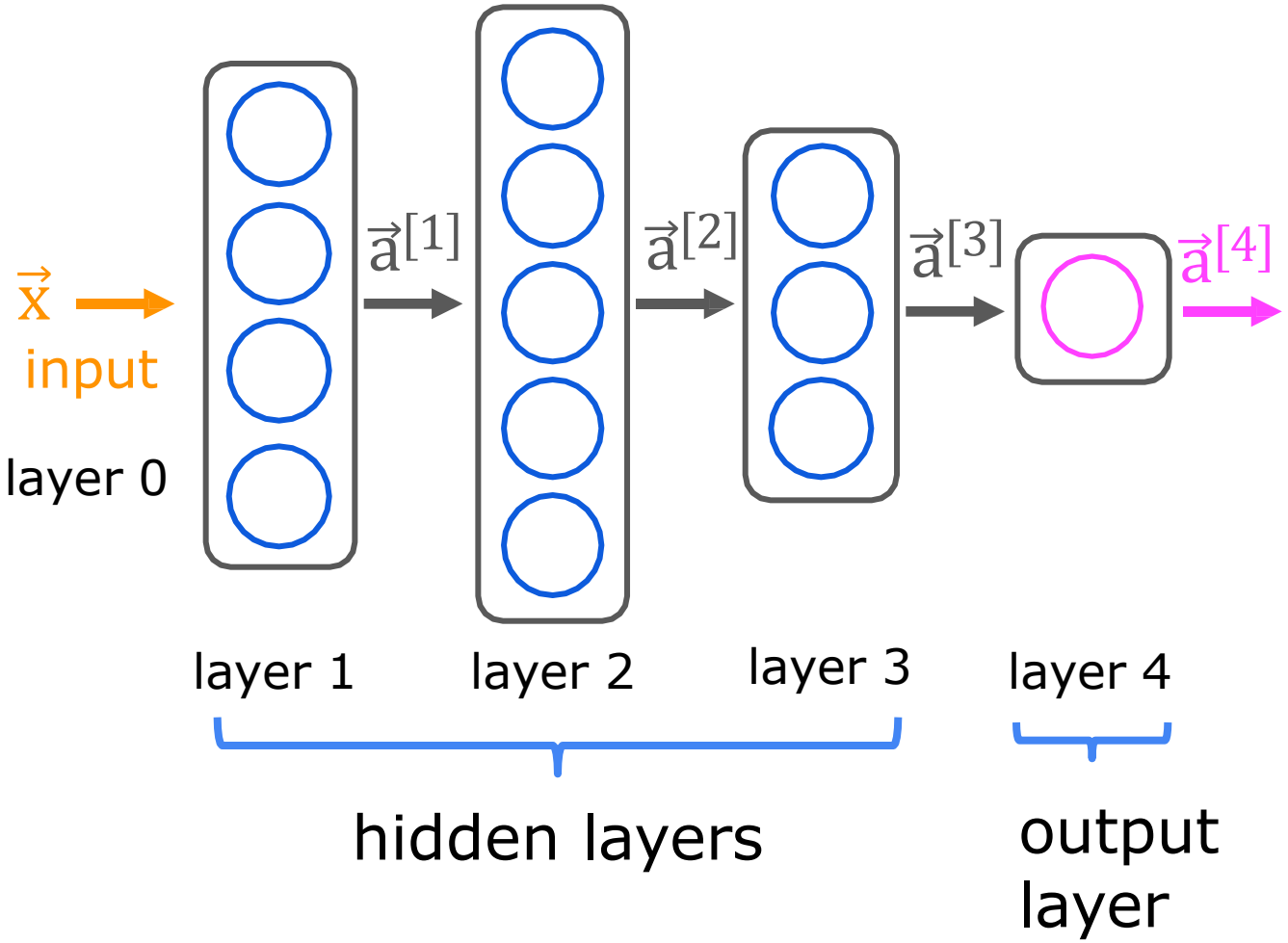
More complex neural network



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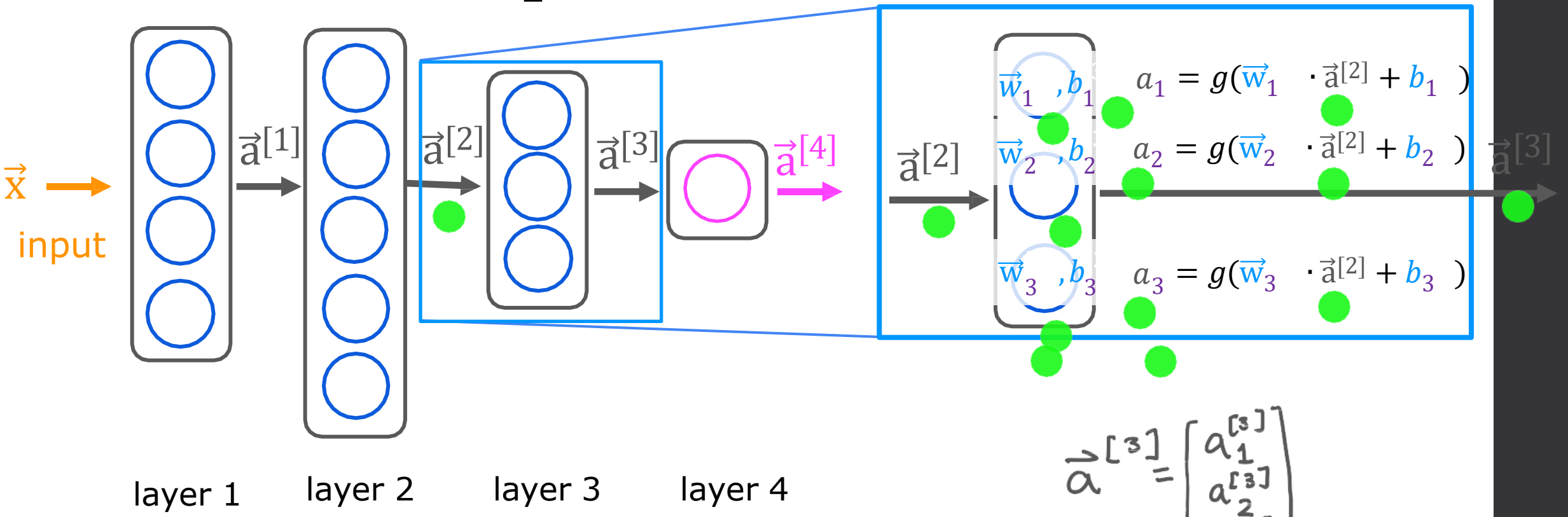
More complex neural network



$$\vec{a}^{[3]} = \begin{bmatrix} a_1^{[3]} \\ a_2^{[3]} \\ a_3^{[3]} \end{bmatrix}$$

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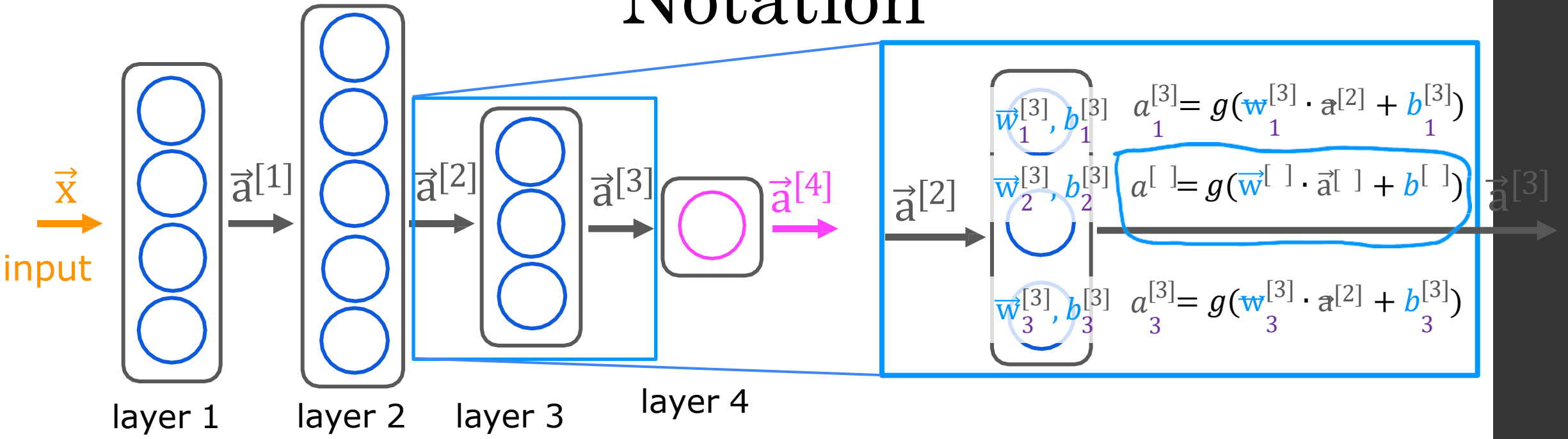
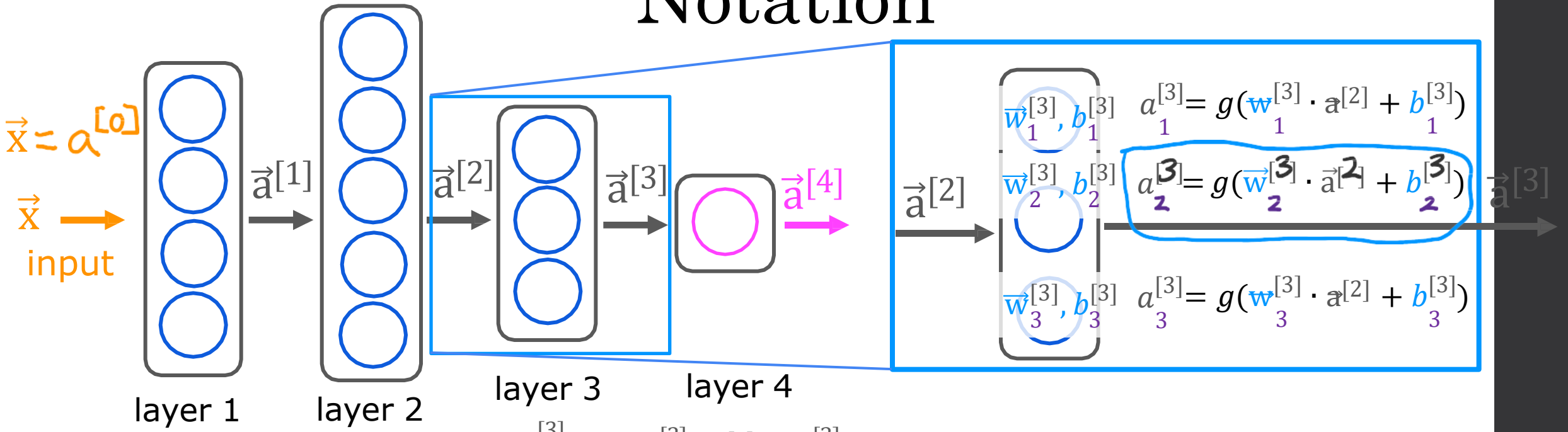
Notation



Question:
Can you fill in the superscripts and subscripts for the second neuron?

Notation



$$a_2^{[3]} = g(\vec{w}_2^{[3]} \cdot \vec{a}^{[2]} + b_2^{[3]})$$

Activation value of layer l, unit(neuron) j

output of layer $l-1$ (previous layer)

$$\boxed{a_j^{[l]} = g(\vec{w}_j^{[l]} \cdot \vec{a}^{[l-1]} + b_j^{[l]})}$$

sigmoid

"activation function"

Parameters w & b of layer l, unit j

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AI

ANI

(artificial narrow intelligence)

E.g., smart speaker,
self-driving car, web search,
AI in farming and factories

AGI

(artificial general intelligence)

Do anything a human can do

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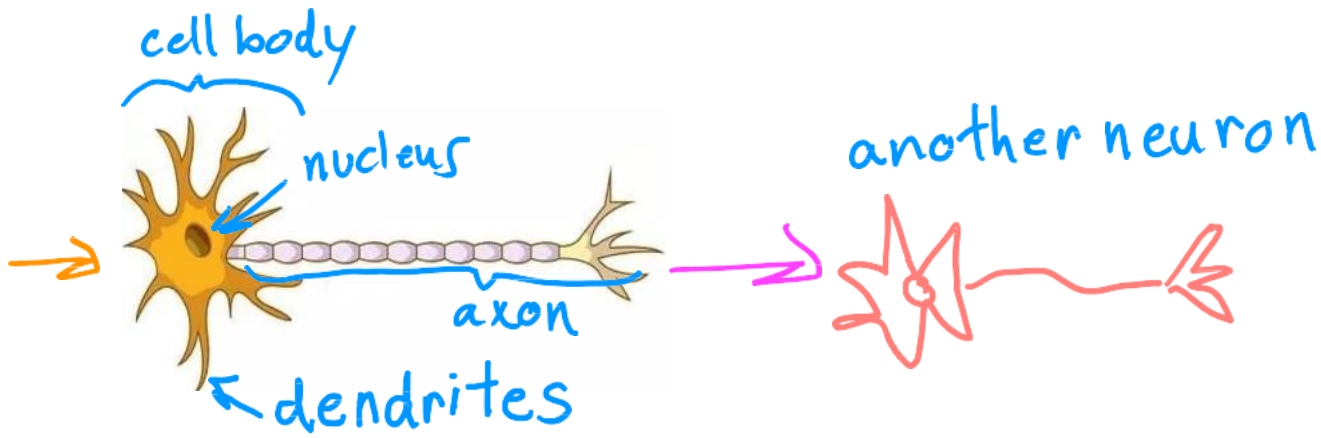
Biological neuron

inputs outputs



Simplified mathematical model of a neuron

inputs outputs

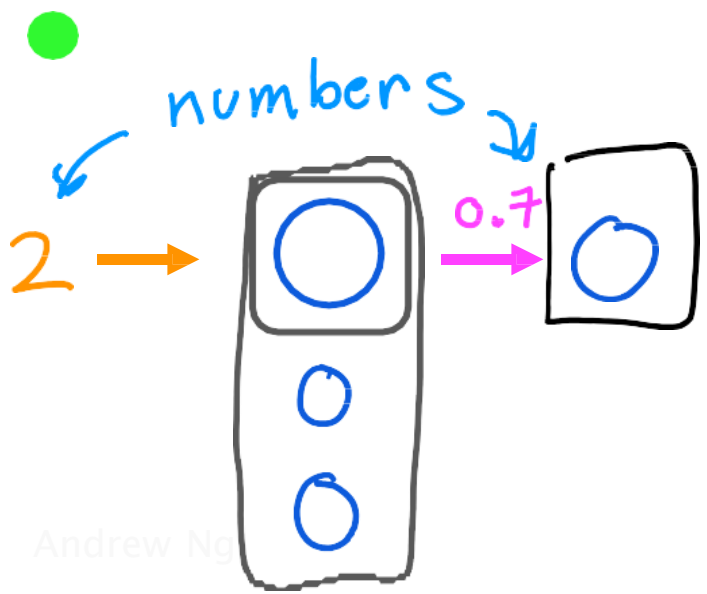
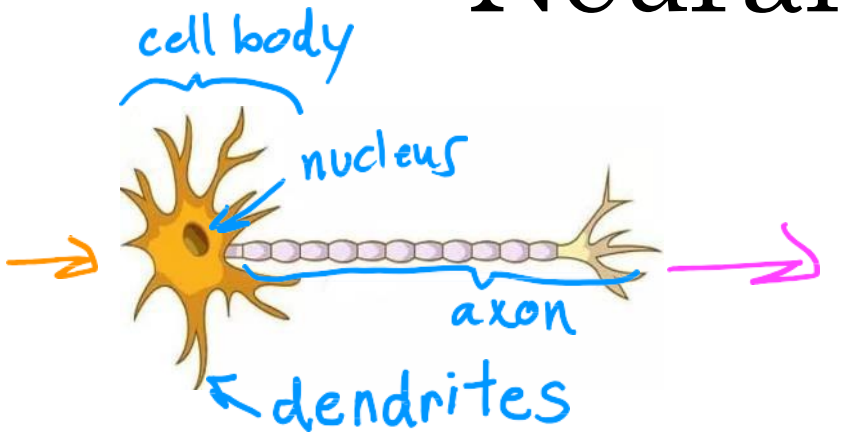


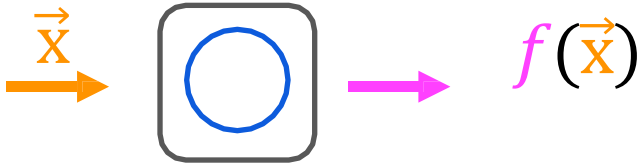
image source: https://biologydictionary.net/sensory-neuron/

Can we mimic the human brain?

Neural network and the brain



vs



We have (almost) no idea how the brain works

Thank You